

ECON 527: Problem Set 4

Instructions

You may collaborate with up to 2 other students to complete the following problems. However, each student must write their own solutions. If you do collaborate with other students, please indicate who you collaborated with on your problem set. In order to receive credit, you must show your work (including documenting your code). Code will be graded both on how well it runs and how well it is organized and documented.

Submission guidelines: please create a single PDF with results and a brief description of your code. Submit your problem set as a compressed (zip) file that contains the PDF and all codes and has the format “PS4_lastname.”

Wealth Distribution in Standard IM Models

In this problem set, we will compare the wealth distribution in standard incomplete market models with the data.

1. Write a function, call it `f_spe` (“stationary partial equilibrium”), that takes as its arguments

- An interest rate r
- A Markov chain for income $(\mathbf{Z}, \mathbf{\Pi})$
- A utility function u , the inverse of its derivative u'^{-1} , a discount factor β , a wealth grid a_i , and a fine wealth grid for computing the stationary distribution of state variables
- Any other parameters you need

and returns

- The value function $V(a_i, z_j)$
- The policy function $g(a_i, z_j)$
- The stationary density of state variables $\phi(a_i, z_j)$

- The net demand for assets $\sum_i \sum_j a_i \phi(a_i, z_j)$

The policy function should be computed using the endogenous gridpoint method with cubic spline interpolation. The distribution of state variables should be computed using the eigenvalue method, with a warning given if the distribution is not unique. The goal is to have a function that can be used in any incomplete market model of the general form described in Lecture 7. Write another function `f_nad` (“net asset demand”) that has the same inputs as `f_spe` and itself calls `f_spe`, but returns only the net demand for assets (this will be useful for finding equilibrium, as many optimization routines require functions that return only one variable).

2. First, let’s compute equilibrium in a Hugget model. Use the flow utility function

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

where $\gamma = 2$. Set the discount factor to $\beta = 0.96$ and assume log earning follows an AR(1) process

$$\log z_t = \rho \log z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

where $\rho = 0.9$ and $\sigma = 0.2$. Discretize income using Rouwenhorst’s method with 7 gridpoints. Set the borrowing limit to average income (note that our calibration corresponds to annual data) and the upper bound to 50 times average income. Discretize wealth using a polynomial grid with curvature parameter 2 and 150 gridpoints. Use the same grid to compute the distribution of state variables.

- (a) Compute the equilibrium interest rate
 - (b) Plot the value function, policy function, and density of state variables. Is the upper bound of the wealth grid binding in equilibrium?
3. Repeat (2) for an Aiyagari model with capital depreciation rate $\delta = 0.10$ and production function

$$F(K, L) = K^\alpha L^{1-\alpha}$$

where $\alpha = 0.36$. Set the borrowing limit to 0 and the upper bound of the asset grid to 60. For all other parameters, use the values from above (the AR(1) process now applies to individual log labor supply instead of earning). In addition, compute the Gini coefficient for wealth and the top 10% and 1% wealth shares (why didn’t we do this for the Huggett model?). How do these compare with the values from the 2019 Survey of Consumer Finances, estimated by Kuhn and Rios-Rull (2020)?

Statistic	2019 SCF
Gini coefficient	0.85
Top 10% share	64.6%
Top 1% share	29.6%

4. Now let's examine the excess demand functions of these models. For the Huggett model (2), plot the excess demand function evaluated on an evenly spaced grid of 25 interest rates over the range $[-0.02, 0.02]$. For the Aiyagari model (3), plot the excess demand function over an evenly spaced grid of 25 interest rates over the range $[-0.02, 0.04]$. The excess demand function in this case is

$$S(r) - K(r)$$

where S denotes aggregate saving and $K(r)$ is the capital stock implied by r .